

APPENDIX C — CONSENT, SHARING FLOWS, AND ACCESS CONTROL MECHANISMS (NON-LIMITING)

A. Overview

This appendix provides non-limiting examples of consent flows, access controls, and sharing mechanisms that may be used to enable cross-device correlation and cross-user correlation while maintaining privacy-preserving outputs. Any mechanism described herein may be substituted or combined with other mechanisms described in the specification.

B. Consent States and Scopes (Non-Limiting)

In various embodiments, the system defines consent as one or more states and/or scopes, including: 1. No Sharing (default), in which only within-user processing occurs. 2. Within-User Sharing Across Nodes, enabling correlation between nodes owned by the same user account. 3. Cross-User Sharing (One-Way), enabling a first user to share selected derived metrics with a second user. 4. Cross-User Sharing (Two-Way), enabling reciprocal exchange of selected derived metrics. 5. Time-Limited Sharing, enabling sharing only for a defined duration. 6. Session-Limited Sharing, enabling sharing only for a defined session or defined data window. 7. Metric-Scoped Sharing, enabling sharing only of selected metric categories (e.g., alignment indices, synchrony curves, confidence-only views). In some embodiments, users select which nodes and which metric categories are permitted for correlation, and permissions may be modified or revoked.

C. Pairing, Invitations, and Verification (Non-Limiting)

Cross-user sharing may be initiated through a pairing flow including one or more of: invitation codes, QR pairing, account linking, mutual confirmation prompts, and/or cryptographic challenges. In certain embodiments, pairing requires confirmation on both user devices. In some embodiments, a pairing is verified using time-limited tokens and device-to-device checks.

D. Access Tokens, Keying, and Secure Exchange (Non-Limiting)

In various embodiments, sharing uses access tokens and/or cryptographic keys, including: 1. Ephemeral tokens that expire automatically, 2. Revocable tokens that can be revoked by either user, 3. Scope-bound tokens that limit which metrics may be exchanged, and 4. Device-bound tokens that bind exchange to authorized devices. In some embodiments, tokens are stored locally on a user device and are encrypted at rest. In certain embodiments, derived metrics are exchanged using encrypted transport and authenticated endpoints.

E. Derived-Metric Sharing Without Raw Signal Sharing (Non-Limiting)

In various embodiments, cross-user correlation is performed without sharing raw sensor signals. Instead, one or more of the following are exchanged: normalized indices, zone-based summaries, quantized curves, embeddings derived from features, confidence indicators, and/or context tags. In certain embodiments, the system performs on-device transformations to produce a privacy-preserving representation prior to any transmission.

F. Revocation, Expiration, and Auditability (Non-Limiting)

In various embodiments: 1. Either user may revoke sharing at any time. 2. Sharing sessions may expire automatically based on time or inactivity. 3. The system may present a sharing status indicator and an access history view. 4. The system may store non-sensitive audit metadata indicating that correlation occurred and which metric categories were used, without storing or exposing raw signals.

G. Safety and Misuse Resistance (Non-Limiting)

In various embodiments, the system incorporates misuse resistance including: minimum data quality thresholds before generating shareable outputs, rate limiting of repeated correlation requests, anomaly detection for spoof attempts, and context validation requiring stable contact integrity and motion gating for eligible windows.

H. Variations

Any consent and sharing flow described herein may be implemented locally, via a mobile application, via a server environment with access controls, or in a hybrid mode. The system may support different privacy modes, including local-only, derived-metrics-only, and consent-gated exchange, while maintaining privacy-preserving outputs.